Clustered Randomized Smoothing for Stochastic Prediction Functions

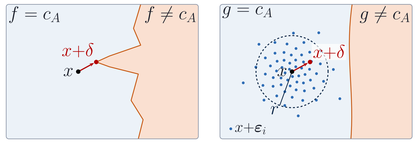
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Randomized smoothing and where it breaks

Randomized smoothing [3]

A classifier f is vulnerable to **adversarial attacks**: a small perturbation δ can flip its prediction (left). Randomized smoothing is provably robust (right): classify under noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $g(x) = \arg\max_c \mathbb{P}\big(f(x+\varepsilon) = c\big)$.

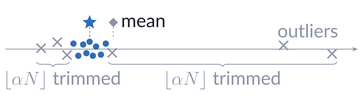


Certificate. If $\mathbb{P}(f(x+\varepsilon) = c_A) \geq \check{p}_A$ and $\max_{c \neq c_A} \mathbb{P}(f(x+\varepsilon) = c) \leq \hat{p}_B$, then

$$g(x+\delta) = c_A \ \text{ for all } \ \|\delta\|_2 < r,$$
$$r = \tfrac{\sigma}{2}\big(\Phi^{-1}(\check{p}_A) - \Phi^{-1}(\hat{p}_B)\big).$$

Randomized smoothing for regression

The natural extension to a stochastic regressor $h_{\mathbf{w}}$ is the expectation $\mathbb{E}_\varepsilon\big[h_{\mathbf{w}}(x+\varepsilon)\big]$. But guarantees are hard to derive: a single **outlier** can push the average out of any set $\mathcal{R}$. [1] proposes α**-smoothing**, which trims the outliers.



α-smoothing [1]:

$$\widetilde{\mathcal{H}}_{N,\alpha}(x) = \frac{1}{N - 2\lfloor \alpha N \rfloor} \sum_{i=\lfloor \alpha N \rfloor + 1}^{N - \lfloor \alpha N \rfloor} \mathcal{H}(x)_{(i)}$$

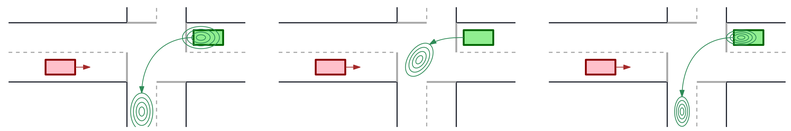
Certificate. For a box $\mathcal{R}$ and all $\|\delta\|_2 \leq r$,

$$\mathbb{P}\big(\widetilde{\mathcal{H}}_{N,\alpha}(x+\delta) \in \mathcal{R}\big) \ \geq \ \sum_{j=N-\lfloor \alpha N \rfloor}^{N} f_{N,j}(\underline{p}_{\mathcal{R}}), \qquad \underline{p}_{\mathcal{R}} = \Phi\Big(\Phi^{-1}(\check{p}_{\mathcal{R}}) - \tfrac{r}{\sigma}\Big),$$

where $\check{p}_{\mathcal{R}} \leq \mathbb{P}\big(h_{\mathbf{w}}(x+\varepsilon) \in \mathcal{R}\big)$ and $f_{n,k}(p) = \binom{n}{k} p^k (1-p)^{n-k}$.

Problem: α-smoothing collapses the modes

Stochastic predictors output **multi-modal** distributions, e.g. a driver can *turn now* or *wait*. In some cases, α-smoothing may lead to **mode collapsing**.



(a) base predictor **(b)** α-smoothing **(c)** clustered α-smoothing

In (b) the output lies between the modes and matches neither behaviour. In (c) **each mode is preserved**.

Clustered α-smoothing: method and guarantee

Method: smooth each mode, then mix

Take a partition $\boldsymbol{\mathcal{V}} = \{\mathcal{V}_1, \ldots, \mathcal{V}_M\}$ of $\mathcal{Y}$, one cell per mode. Let $I_m = \{\, i : h_{\mathbf{w}}(x+\varepsilon_i) \in \mathcal{V}_m \,\}$ and $\mathcal{H}_{\mathcal{V}_m}(x) = \big(h_{\mathbf{w}}(x+\varepsilon_i)\big)_{i \in I_m}$.

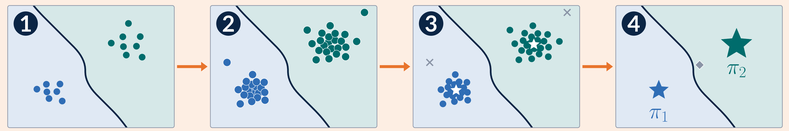
(i) Trim inside each cluster:

$$\widetilde{\mathcal{H}}_{N,\alpha,\mathcal{V}_m}(x) = \frac{1}{|I_m| - 2\lfloor \alpha |I_m| \rfloor} \sum_{i=\lfloor \alpha |I_m| \rfloor + 1}^{|I_m| - \lfloor \alpha |I_m| \rfloor} \mathcal{H}_{\mathcal{V}_m}(x)_{(i)}$$

(ii) Mix the clusters:

$$\widetilde{\mathcal{H}}_{N,\alpha,\boldsymbol{\nu}}(x) = \sum_{m=1}^{M} \mathbf{1}_{\mathbf{z}=m}\, \widetilde{\mathcal{H}}_{N,\alpha,\mathcal{V}_m}(x), \qquad \mathbb{P}(\mathbf{z} = m) = \pi_m = \tfrac{|I_m|}{N}.$$

Note that: i) if $M = 1$ ($\boldsymbol{\mathcal{V}} = \{\mathcal{Y}\}$), we retrieve α-smoothing [1]; ii) $\alpha = 0$ gives RS-Reg [2].



Partition — $\bar{N}$ samples, clustered Voronoi cells $\mathcal{V}_m$

Sample — N new samples fall in the cells

Trim per cluster — $\times$ trimmed, $\star$ trimmed mean

Mix — pick m w.p. π_m, $\blacklozenge$ = α-smoothing

Theory: robustness certificate (Theorem 1)

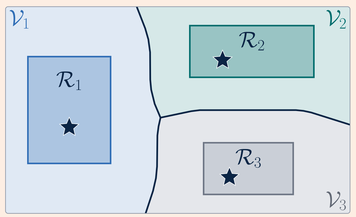
Let $\mathcal{R}_l \subset \mathcal{V}_l$, $l \in \mathcal{L} \subseteq [M]$, be **axis-aligned boxes** and $\tilde{\mathcal{R}} = \bigcup_{l \in \mathcal{L}} \mathcal{R}_l$. **Assume** $\check{p}_{\mathcal{V}_m} \leq \mathbb{P}_{(\mathbf{w},\varepsilon)}\big(h_{\mathbf{w}}(x+\varepsilon) \in \mathcal{V}_m\big) \leq \hat{p}_{\mathcal{V}_m}$ and $\mathbb{P}_{(\mathbf{w},\varepsilon)}\big(h_{\mathbf{w}}(x+\varepsilon) \in \mathcal{R}_l\big) \geq \check{p}_{\mathcal{R}_l}$. For $r \geq 0$, let

$$\underline{p}_{\mathcal{V}_m} = \Phi\Big(\Phi^{-1}(\check{p}_{\mathcal{V}_m}) - \tfrac{r}{\sigma}\Big), \ \ \overline{p}_{\mathcal{V}_m} = \Phi\Big(\Phi^{-1}(\hat{p}_{\mathcal{V}_m}) + \tfrac{r}{\sigma}\Big), \ \ \underline{p}_{\mathcal{R}_l} = \Phi\Big(\Phi^{-1}(\check{p}_{\mathcal{R}_l}) - \tfrac{r}{\sigma}\Big).$$

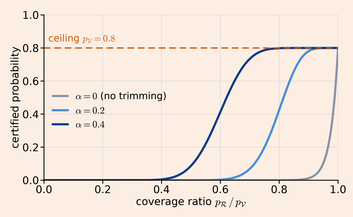
Then for all $\|\delta\|_2 \leq r$,

$$\mathbb{P}_{(\mathbf{w},\varepsilon_1,\ldots,\varepsilon_N)}\big(\widetilde{\mathcal{H}}_{N,\alpha,\boldsymbol{\nu}}(x+\delta) \in \tilde{\mathcal{R}}\big) \ \geq \ \inf_{p \in \mathcal{A}} \sum_{l \in \mathcal{L}} \sum_{s=1}^{N} \underbrace{\tfrac{s}{N}}_{\text{cluster } l \text{ is drawn}} \underbrace{f_{N,s}(p_{\mathcal{V}_l})}_{s \text{ samples fall in } \mathcal{V}_l} \underbrace{\sum_{j=s-\lfloor \alpha s \rfloor}^{s} f_{s,j}\Big(\tfrac{\underline{p}_{\mathcal{R}_l}}{p_{\mathcal{V}_l}}\Big)}_{\geq s-\lfloor \alpha s \rfloor \text{ of them fall in } \mathcal{R}_l}$$

with $\mathcal{A} = \big\{p : p_{\mathcal{V}_m} \in [\underline{p}_{\mathcal{V}_m}, \overline{p}_{\mathcal{V}_m}],\ \sum_m p_{\mathcal{V}_m} = 1\big\}$.



(a) Generalized Voronoi partition of the boxes $\mathcal{R}_l$ $\star$ = smoothed output.

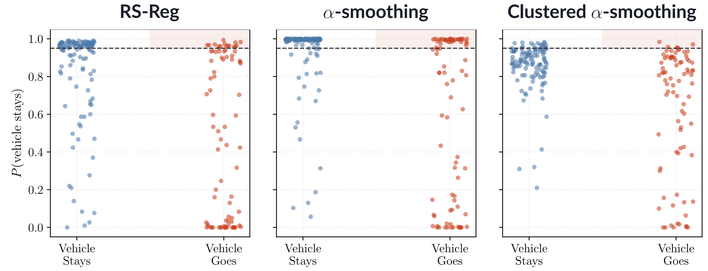


(b) Bound vs. coverage ratio (N=50, $p_{\mathcal{Y}}$=0.8).

Experimental results

Experiment 1: trajectory prediction

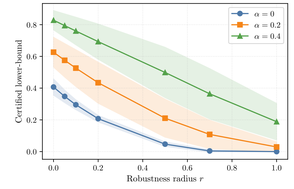
TrajFlow [4] predicts 2 s of a driver's trajectory, $h_{\mathbf{w}} : \mathbb{R}^{18} \to \mathbb{R}^{40}$, on L-GAP [5]: drivers *go* or *stay*. 200 random inputs, one cluster per behaviour. **Safety rule:** ego goes if $\mathbb{P}(\text{other stays}) > 95\%$.



One dot per input: predicted $\mathbb{P}(\text{other stays})$; blue = it stayed, red = it went. Red dots above the dashed line are **risky decisions**.



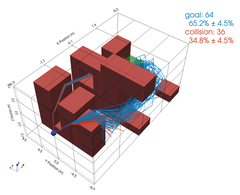




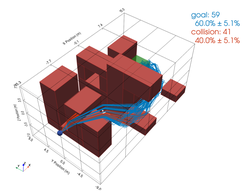
Middle: mean $\mathbb{W}_2$ over 200 inputs in $\mathbb{R}^{40}$ (std. 2.09 / 2.19 / 1.59, in bar order). Right: certified bound of Theorem 1 vs. radius r for 20 inputs (mean ± 1 std., $p = 0.85$, $\alpha \in \{0, 0.2, 0.4\}$).

Experiment 2: multi-modal quadrotor control

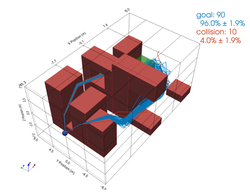
Benchmark [6]: a PPO policy with a 2-component Gaussian-mixture action distribution, two gaps lead to the goal. Parameters: $\sigma = 0.05$, $N = 30$, $\alpha = 0.2$, DBSCAN, coverage $p = 0.9$, $\beta = 0.06$.



nominal policy
collision 34.8% ± 4.5%



α-smoothing
collision 40.0% ± 5.1%



clustered α-smoothing
collision 4.0% ± 1.9%

100 rollouts per policy (mean ± std. over 100 simulations).

Key takeaways

1. **Mode collapse.** α-smoothing may average the modes, while clustered α-smoothing keeps them as a mixture.
2. **Guarantee.** Theorem 1 extends the protection against adversarial attacks to clustered α-smoothing.
3. **Smoothed regressor remains safe.** Risky decisions $16.5\% \to 2.5\%$, collisions $40.0\% \to 4.0\%$ vs. α-smoothing.

References. [1] Rekavandi, Farokhi, Ohrimenko, Rubinstein. Certified adversarial robustness via randomized α-smoothing for regression models. NeurIPS 2024. [2] Rekavandi, Ohrimenko, Rubinstein. RS-Reg. TMLR 2025. [3] Cohen, Rosenfeld, Kolter. Certified adversarial robustness via randomized smoothing. ICML 2019. [4] Mészáros, Schumann, Alonso-Mora, Zgonnikov, Kober. TrajFlow. IEEE IV 2024. [5] Zgonnikov, Abbink, Markkula. Should I stay or should I go? Human Factors 2024. [6] Badings et al. Robust control for dynamical systems with non-Gaussian noise via formal abstractions. JAIR 2023. Supported by NWO (OCENW.M.22.056) and EPSRC (EP/Y028872/1).